

Empirical Finance: Methods and Applications

Group Coursework 1

Group 3

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Executive Summary

As a team that has been hired by Imperial Global Asset Management, we look to analyse the US and UK stock market index, relationships with their respective treasury yields and identify if there is a strategy to maximize wealth based on these factors. For our exercises, we utilise over 100 years of data (from 1920 to 2026) in order to have a robust sample size for data analysis.

In Exercise 1, we dive deep into US data, specifically the S&P 500 (US stock market index), short-term yield, long-term yield, and exchange rate. We analyse their summary statistics, ACF and PACF, as well as time-series analyses to have a deep statistical understanding of our data in order to identify profitable portfolios or trading strategies. In Exercise 2, we investigate the effectiveness of the US term spread as a predictor for the S&P 500 returns by comparing it to a benchmark model, analysing an out-of-sample R^2 , running the Clark and West test, and performing a bootstrapping test. Finally, in Exercise 3, we take the perspective of a US-based investor and can allocate our wealth among the following assets: US risk-free asset, US stock market, and the UK stock market. We perform analysis on portfolio performance rebalancing by looking at the Sharpe Ratio, performance fee, break-even transaction costs and volatility. We then provide an economic recommendation based on the results.

Exercise 1

1.1 Data Preparation and Variable Construction

When calculating return, there are two metrics: while simple returns are highly intuitive, log returns are preferred for time-series analysis due to their time additivity, which is an essential property when calculating returns over multiple periods. Furthermore, log returns are closer to a normal distribution and maintain stationarity, which are critical requirements for ACF and PACF analysis and parameter estimation. Therefore, we use log returns. Let P_t denotes the index value at time t and S_t denotes the GBPUSD exchange rate:

$$r_t^{US} = \ln\left(\frac{P_t}{P_{t-1}}\right) = \ln(P_t) - \ln(P_{t-1}) \quad (1)$$

$$r_t^{FX} = \ln\left(\frac{S_t}{S_{t-1}}\right) = \ln(S_t) - \ln(S_{t-1}) \quad (2)$$

In addition to stock and exchange rate returns, we also consider changes in US Treasury yields. Let y_t^{short} and y_t^{long} denote the short-term and long-term US Treasury yields, respectively. Since US Treasury yields are known to be highly persistent and often exhibit near unit-root behavior in levels,

we take first differences to help reduce persistence and improve stationarity.

$$\Delta y_t^{\text{short}} = y_t^{\text{short}} - y_{t-1}^{\text{short}}, \quad (3)$$

$$\Delta y_t^{\text{long}} = y_t^{\text{long}} - y_{t-1}^{\text{long}}. \quad (4)$$

1.2 Summary Statistics and Data Description

Before formal statistical summaries and time series diagnostics, Figure 1 plots the time series of US stock returns and the GBP/USD exchange rate returns. Visualizing the data provides an intuitive understanding of their volatility, persistence, and potential structural features, which is useful for the summary statistics and autocorrelation analysis.

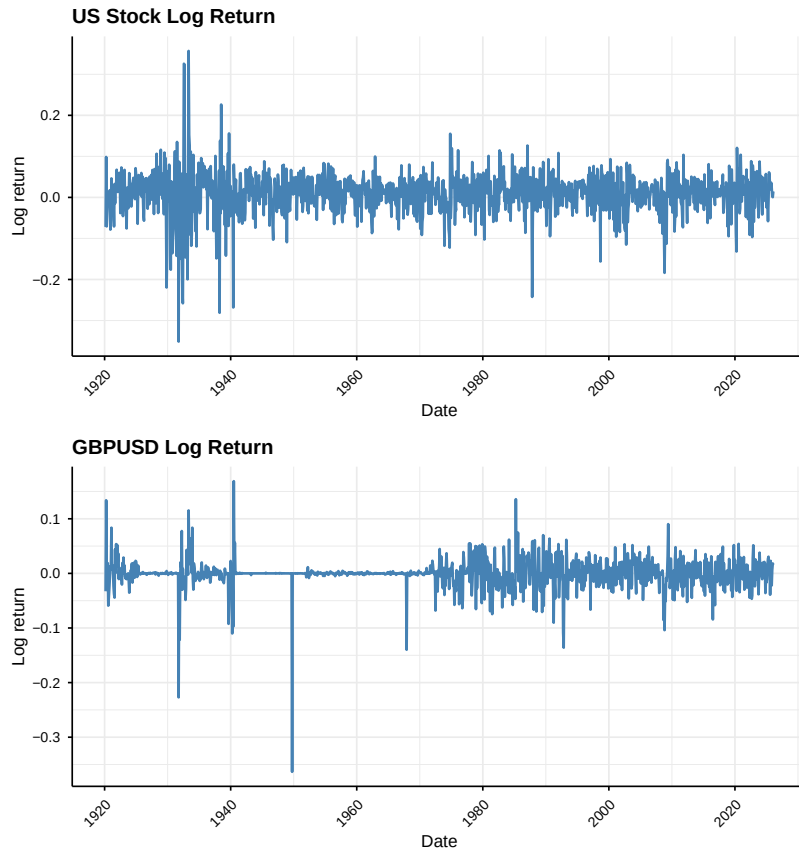


Figure 1: Time series of US stock returns and GBP/USD exchange rate returns

Tables 1 and 2 report the summary statistics for US stock log returns, GBP/USD log exchange rate returns, and delta US Treasury yields. All series are based on a common sample of 1,272 monthly observations. Table 1 summarizes the mean, median, standard deviation, min, max and first-order serial correlation of the series, while Table 2 shows both conventional and robust measures of skewness and kurtosis.

As reported in Table 1, the first-order serial correlation of stock and exchange rate returns remains

close to zero, suggesting limited linear predictability at the monthly frequency. For both US stock returns and GBP/USD exchange rate returns, the mean is small relative to the standard deviation. In particular, the mean return of the exchange rate series is close to zero, whereas stock returns exhibit substantially higher volatility.

As shown in Table 2, both return series display pronounced departures from normality. Returns are negatively skewed and exhibit substantial excess kurtosis, indicating asymmetric and heavy-tailed distributions. This suggests that large negative return and extreme movements occurred more frequently than implied by a normal distribution. These findings are reinforced by robust kurtosis measures, such as the Moor's and Crow & Siddiqui's kurtosis, indicating that the heavy-tailed behavior is not driven by a small number of extreme outliers. While yield changes remain non-normal and exhibit excess kurtosis, their serial dependence is limited.

Table 1: Core summary statistics

	N	Mean	SD	Min	Q05	Median	Q95	Max	AC1
US_Stock_Return	1272	0.834	5.270	-35.136	-7.599	1.277	7.651	35.693	0.064
GBPUSD_Return	1272	-0.074	2.669	-36.340	-3.888	0.000	3.742	16.887	0.055
Delta_US_Short_Yield	1272	-0.001	0.384	-3.850	-0.499	0.000	0.430	2.400	0.126
Delta_US_Long_Yield	1272	-0.001	0.257	-1.880	-0.380	0.000	0.400	1.590	0.111

Table 2: Skewness and kurtosis measures, including robust statistics

	Skewness	Kurtosis_Excess	Skewness_Bowley	Skewness_GM	Kurtosis_Moors	Kurtosis_CS
US_Stock_Return	-0.511	7.725	-0.070	-0.122	0.130	0.668
GBPUSD_Return	-2.456	34.871	-0.057	-0.052	2.178	5.892
Delta_US_Short_Yield	-1.553	20.750	0.294	-0.003	1.241	5.659
Delta_US_Long_Yield	-0.318	7.024	0.000	-0.003	0.570	2.463

1.3 ACF and PACF Analysis

Figures 2 present the sample autocorrelation functions (ACF) and partial autocorrelation functions (PACF) for US stock log returns and GBP/USD log exchange rate returns. The ACF and PACF are plotted up to 24 lags.

For US stock log returns, The ACF indicates that autocorrelations decay rapidly and remain largely insignificant across lags, while the PACF does not reveal a clear cutoff or systematic structure. A similar pattern is observed for GBP/USD log exchange rate returns. These results suggest limited linear predictability in both stock and exchange rate returns at the monthly frequency when using past returns as predictors.

Figure 3 presents the ACF and PACF for changes in US short-term and long-term Treasury yields up to 24 lags. For both changes in short-term and long-term Treasury yields, the ACF exhibits a small but positive autocorrelation at the first lag, followed by a rapid decay towards zero. Similarly, their

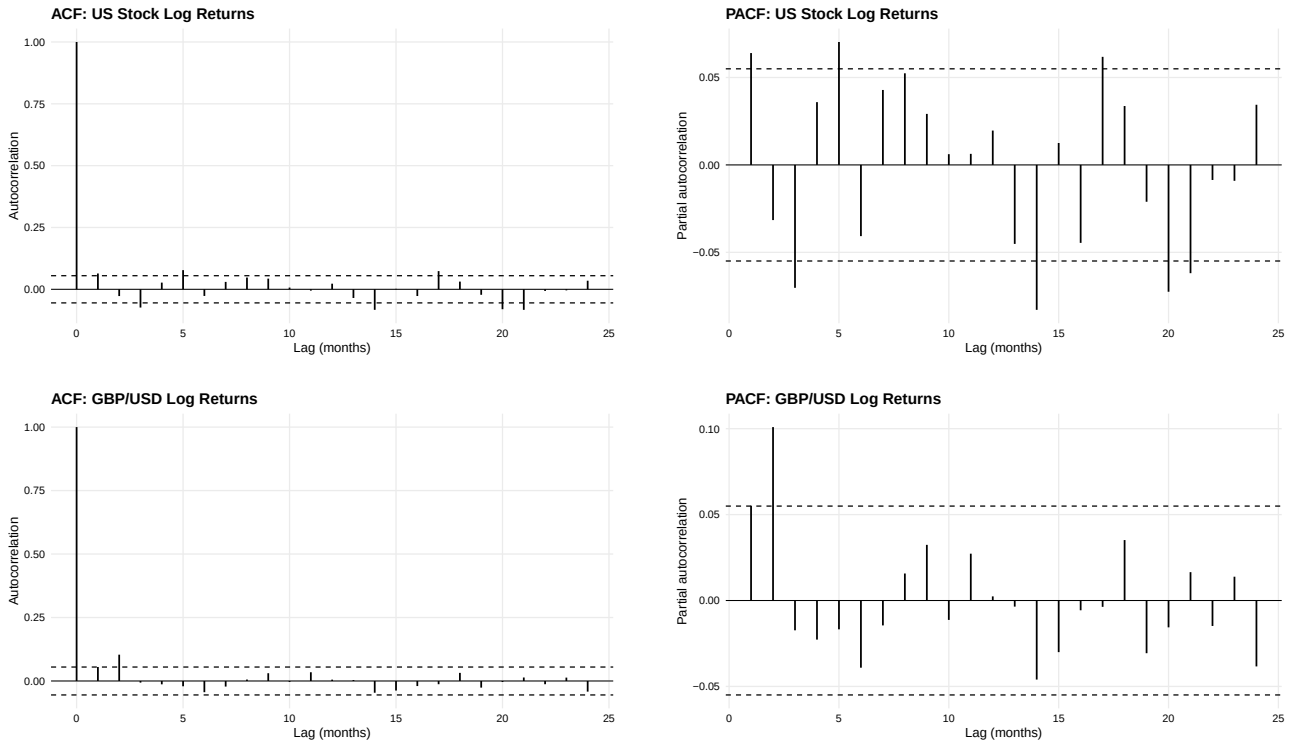


Figure 2: ACF and PACF of US stock and GBP/USD log returns

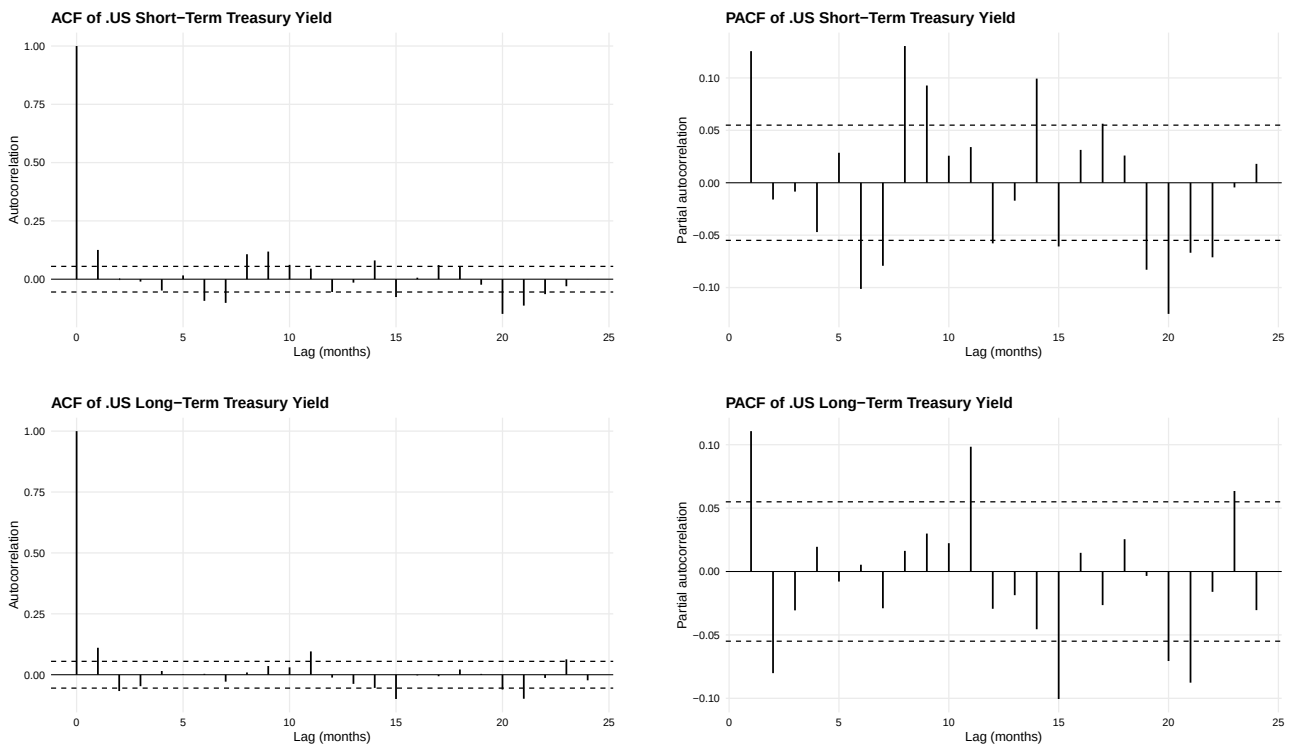


Figure 3: ACF and PACF of changes in US short-term and long-term Treasury yields

Table 3: Selected ARMA specifications and maximum-likelihood estimates

Model	Param	Estimate	SE	t	AICc	BIC
US stock: ARMA(0,0)	intercept	0.0083	0.0015	5.6429	-3874.7698	-3864.4825
GBP/USD: ARMA(0,2)	ma1	0.0509	0.0279	1.8251	-5618.5888	-5598.0270
GBP/USD: ARMA(0,2)	ma2	0.1086	0.0281	3.8659	-5618.5888	-5598.0270
GBP/USD: ARMA(0,2)	intercept	-0.0007	0.0009	-0.8532	-5618.5888	-5598.0270
Delta US short yield: ARMA(0,1)	ma1	0.1274	0.0278	4.5770	1157.9031	1173.3292
Delta US short yield: ARMA(0,1)	intercept	-0.0007	0.0120	-0.0542	1157.9031	1173.3292
Delta US long yield: ARMA(0,1)	ma1	0.1270	0.0292	4.3431	138.6777	154.1038
Delta US long yield: ARMA(0,1)	intercept	-0.0005	0.0081	-0.0649	138.6777	154.1038

PACF does not reveal a clear cutoff or systematic autoregressive structure, indicating that higher-order lagged effects are weak. Overall, Figure 3 suggests that changes in US Treasury yields exhibit only limited and short-run serial dependence rather than strong persistence.

1.4 Model Specification and Estimation

To model the dynamic properties of the monthly time series, we consider an ARMA(p, q) specification for each variable of interest. Let x_t denote a generic stationary series, including US stock log returns, GBP/USD log exchange rate returns, and changes in US Treasury yields. The ARMA(p, q) model is given by:

$$x_t = c + \sum_{i=1}^p \phi_i x_{t-i} + \varepsilon_t + \sum_{j=1}^q \theta_j \varepsilon_{t-j}, \quad \varepsilon_t \sim \text{i.i.d. } (0, \sigma^2), \quad (5)$$

where ϕ_i and θ_j are the autoregressive and moving-average coefficients, respectively, and ε_t is a white-noise innovation.

Table 3 reports the parameter estimates of the selected ARMA models for US stock returns, GBP/USD exchange rate returns, and changes in US Treasury yields.

For US stock log returns, the ARMA(0,0) specification implies that returns follow a white-noise process with a constant mean. The estimated intercept is statistically significant but economically small. This result indicates that while average returns are positive over the sample period, there is no systematic dependence on past shocks or past returns, which is consistent with the absence of significant autocorrelation observed in the ACF and PACF.

For GBP/USD log exchange rate returns, the ARMA(0, 2) model includes two moving-average terms. The estimated MA coefficients are modest in magnitude, with the second lag being statistically significant. This suggests that past exchange rate innovations may affect current returns over a very short horizon, but these effects dissipate quickly. Importantly, the lack of autoregressive terms

indicates that past exchange rate returns themselves do not provide meaningful predictive power for future movements.

Changes in US Treasury yields exhibit a different but still limited form of serial dependence. For both short-term and long-term yield changes, the ARMA(0, 1) specification captures a small but statistically significant moving-average component. The estimated MA(1) coefficients indicate that yield changes respond to recent shocks in the previous period.

Overall, the estimation results indicate that asset returns are only weakly predictable using their own past values at the monthly frequency, while changes in Treasury yields display only slight and short-lived dependence on recent innovations.

Exercise 2

2.1 Economic Motivation of Study

With a robust statistical understanding of the US stock market index (S&P 500), treasury yields, and exchange rate in Exercise 1, we are now interested in forming a prediction model for the returns of the S&P 500 to identify if we can forecast future returns and therefore develop a systematic and profitable trading strategy.

A common predictor for the S&P 500 monthly returns is the term spread (defined as the difference between the 10-year and 3-month US Treasury rates). The term spread has been studied widely and is commonly used as an early warning sign for financial instability (Rudebusch and Williams, 2008). For example, an inverted yield curve is commonly tied to a recession or economic downturn as in normal times, a bond with a longer maturity fetches a higher rate on the market due to increased duration risk. However, in the lead up to an economic downturn, changes in risk attitudes can drive the yield for short-term bond above the yield for long-term bond.

In the case of stock returns, there is a positive correlation of the slope of the yield curve of US treasuries to excess returns on the US stocks (Fama and French, 1989). A positive spread indicates investors positivity about near-term economic prospects which translates to better stock performance. Thus, we will look to analyse the predictive power of the term spread for returns of the S&P 500. To make our backtest robust, we use over 100 years of data, starting from Jan 1920 to Jan 2026 with a 10-year rolling window (120 months) for out-of-sample testing.

2.2 Mathematical Explanations of Predictor Regression

For this exercise, we use a one-step ahead predictive regression, which is a linear specification designed to assess the predictive power of past economic or financial variable for future asset returns and is formulated as:

$$y_t = \alpha + \beta x_{t-1} + \epsilon_t \quad (6)$$

This is our competing model with an initial window of 10 years, where:

- y_t denotes the monthly log return of the S&P 500 between time $t - 1$ and t , with log returns used to ensure stationarity as in Exercise 1;
- x_{t-1} is the term spread at time $t - 1$, used as a lagged predictor;
- β measures the predictive significance of x_{t-1} for y_t ;
- α is the intercept term;
- ϵ_t represents the error term, capturing the component of y_t unexplained by the model;
- $t = 2, \dots, 120$, corresponding to February 1920 to December 1929 for the initial in-sample training period;
- $t - 1 = 1, \dots, 119$, corresponding to January 1920 to November 1929.

Estimating a predictive regression is challenging in finance as returns are generally noisier with fast mean-reverting dynamics, and shocks driving y_t and x_t often display contemporaneous negative correlation. As such, we complement the conventional in-sample analysis with an out-of-sample assessment.

For our analysis, we also use an OLS estimator instead of the LAD estimator since it has a high efficiency, fast and exact computation, and is widely used by hedge funds to test systematic strategies in the real world. Via the OLS, we have:

$$\hat{\beta} = \frac{\text{Cov}(y_{t+1}, x_t)}{\text{Var}(x_t)} \quad (7)$$

$$\hat{\alpha} = \bar{y} - \hat{\beta} \bar{x} \quad (8)$$

which in the matrix form is represented as:

$$\underbrace{\begin{bmatrix} y_2 \\ y_3 \\ \vdots \\ y_{120} \end{bmatrix}}_Y = \underbrace{\begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_{119} \end{bmatrix}}_X \underbrace{\begin{bmatrix} \alpha \\ \beta \end{bmatrix}}_{\beta} + \underbrace{\begin{bmatrix} \epsilon_2 \\ \epsilon_3 \\ \vdots \\ \epsilon_{120} \end{bmatrix}}_{\epsilon}$$

where we let b be the vectors of the OLS coefficients:

$$b = \begin{bmatrix} \hat{\alpha} \\ \hat{\beta} \end{bmatrix} = (X'X)^{-1}(X'Y) \quad (9)$$

We now generate a one-step forecast of the next period return (out-of-sample):

$$\mathbb{E}(y_{t+1}) = \hat{y}_{t+1} = \hat{\alpha} + \hat{\beta}x_t \quad (10)$$

with $t+1 = 121$, corresponding to Jan 1930 and $t = 120$, corresponding to Dec 1929.

Afterwards, we construct the forecast error as:

$$\hat{\epsilon}_{t+1} = y_{t+1} - \hat{y}_{t+1} \quad (11)$$

Lastly, we repeat with an expanded training set (expanding by adding the next period term spread observation until we reach the end of our data, given as follows:

$$y_t = \alpha + \beta x_{t-1} + \epsilon_t \quad (12)$$

Now, in order to assess our out-of-sample predictability, we use a common benchmark model, which is the random walk (nested with $\beta = 0$), given as:

$$y_t = \alpha + \epsilon_t \quad (13)$$

where α is the historical mean of y_t observed at time t .

2.3 Statistical Evaluation: Out-of-sample R^2

To evaluate if our competing model outperforms the benchmark model, we utilise the Mean Squared Error (MSE):

$$\text{MSE} = \frac{1}{T-t} \sum_{i=t+1}^T \hat{\epsilon}_i^2 \quad (14)$$

and then compute the out-of-sample (OOS) R^2 :

$$R_{\text{OOS}}^2 = 1 - \frac{\text{MSE}_{\text{MODEL}}}{\text{MSE}_{\text{BENCHMARK}}} \quad (15)$$

Table 4: Results Summary

Beta	OOS R^2
0.1880	-0.01363

From Table 4, we see that beta (the slope of the competing model) is positive, indicating that the term spread is positively correlated with US market returns (i.e higher spreads tend to correspond to periods of higher S&P 500 returns). From a financial standpoint, this result verify the hypothesis of investors' market expectations for the future driving returns in the US stock market.

However, we see that the OOS R^2 is negative, highlighting that our competing model underperformed compared to the benchmark. Thus, while the term spread is positively correlated to market returns, it cannot be used as a predictor to forecast future market returns accurately.

2.4 Statistical Evaluation: Clark and West Test

While we understand that the competing model underperformed relative to the benchmark, we now implement the Clark and West test to evaluate whether the competing model provides a better out-of-sample forecast accuracy compared to the benchmark model.

$$H_0 : \text{MSE}_{\text{MODEL}} = \text{MSE}_{\text{BENCHMARK}} \quad (\text{Term spread adds no predictive value})$$

$$H_A : \text{MSE}_{\text{MODEL}} < \text{MSE}_{\text{BENCHMARK}} \quad (\text{Term spread improves forecast accuracy})$$

Let $\hat{y}_{t+1,\text{BEN}}$ and $\hat{y}_{t+1,\text{MOD}}$ denote one-step ahead historical mean and term spread forecasts, with forecast errors $\hat{\epsilon}_{t+1} = y_{t+1} - \hat{y}_{t+1}$. The test adjustment accounts for the upward bias in expanded model's

MSE due to parameter estimation noise:

$$f_t = \hat{\epsilon}_{t+1,\text{BEN}}^2 - [\hat{\epsilon}_{t+1,\text{MOD}}^2 - \underbrace{(\hat{y}_{t+1,\text{BEN}} - \hat{y}_{t+1,\text{MOD}})^2}_{\text{adjustment term}}] \quad (16)$$

We compute f_t for $t = t_0, \dots, T - 1$ and regress on a constant:

$$f_i = \rho + \eta_i \quad (17)$$

The CW statistic is:

$$CW = \frac{\hat{\rho}}{se(\hat{\rho})} \quad (18)$$

The Clark and West test evaluates whether the predictive information in the term spread is economically meaningful in an out-of-sample context.

CW Statistic	Critical Value (5%)	Decision
0.486	1.65	$0.486 < 1.65$, fail to reject H_0 at 5% significance level

Table 5: Clark and West Test Results

At the 5% significance level, we fail to reject H_0 . There is insufficient evidence that the term spread provides statistically significant out-of-sample forecasts for log returns than the benchmark model. The benchmark model remains adequate, and the additional complexity of MOD does not yield forecast improvement.

This suggests that while the term spread may exhibit in-sample predictive power, it does not translate to a reliable out-of-sample forecast relative to the simple benchmark.

2.5 Economic Restriction

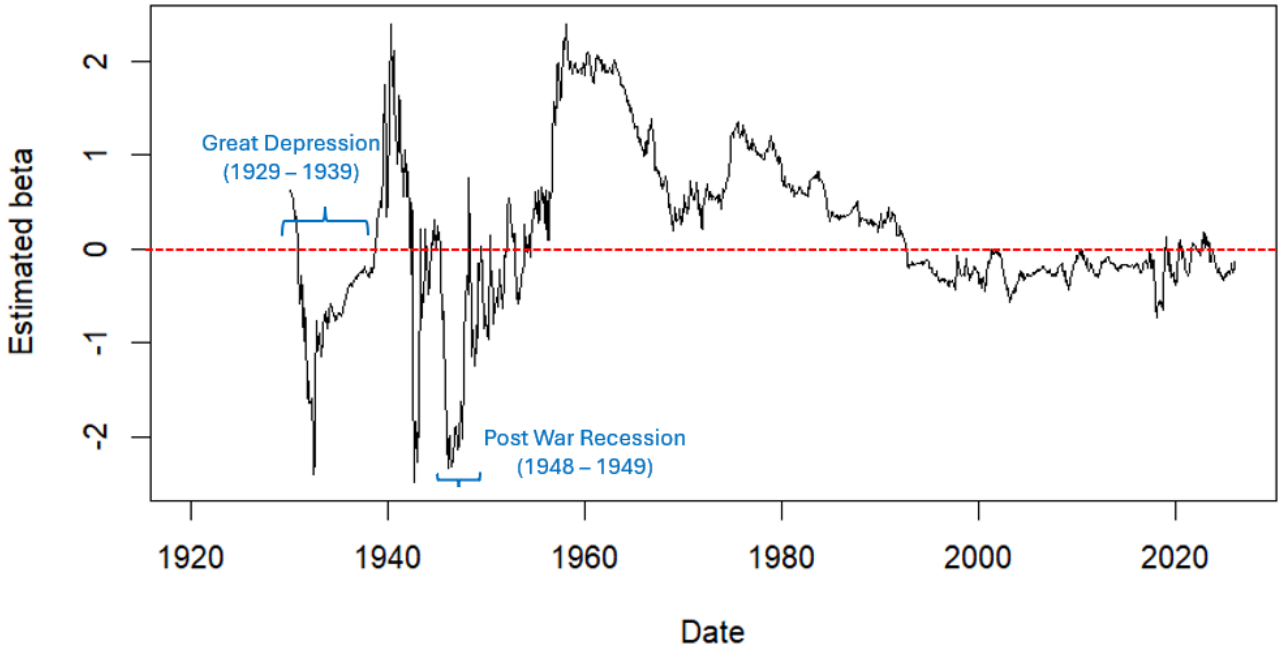


Figure 4: Estimated beta over entire time period of dataset

Throughout our forecast evaluation period (1930-2026), several economic shocks significantly distort the estimated relationship between the term spread and log returns. As shown in Figure 4, beta estimates plunge below -2 during two major crises:

1. The Great Depression: $\hat{\beta}$ experiences a sudden drop beginning in 1930. This is because the initial training window (1920-1929) contains no data in the Depression era, yet the model must forecast into the most severe economic crisis in modern history.
2. The Post-War Recession: Similar experience emerge as the economy transitions from wartime mobilisation to peacetime adjustment.

In both cases, $\hat{\beta}$ gradually recover to positive values as more crisis-period data enters the rolling training window. This pattern reveals a critical limitation of rolling-window OLS: extreme observations during structural breaks produce coefficient estimates that are both economically implausible and statistically unstable, yielding poor R^2_{OOS} and CW results as observed.

From an economic perspective, theory suggests that a high term spread (steep yield curve) should predict higher future economic growth and expected stock returns. This suggests that the relationship between the term spread and future returns should be non-negative i.e. $\hat{\beta} \geq 0$.

The negative beta estimates in Figure 4 contradicts the theory. To address this problem, we implement the sign restriction proposed by Campbell and Thompson (2008):

1. Set $\hat{\beta} = 0$ when $\hat{\beta} < 0$
2. Set $\hat{y}_{t+1} = 0$ when $\hat{y}_{t+1} < 0$
3. Set $\hat{\beta} = 0$ when $\hat{\beta} < 0$ and $\hat{y}_{t+1} = 0$ when $\hat{y}_{t+1} < 0$

where $\hat{y}_{t+1}$ is the excess stock return and $\hat{\beta}$ the slope coefficient.

In our analysis, we forecast log stock returns rather than excess returns. The restrictions on $\hat{y}_{t+1}$ specifically applies to excess returns. Therefore, we only implement the first restriction on $\hat{\beta}$ to ensure that the term spread never receives negative weight in our forecasting model, consistent with the economic interpretation that higher term spreads should predict higher future returns.

We anticipate that imposing this restriction will reduce model sensitivity to outlier observations during economic crises.

Table 6 compares model performance before and after applying the coefficient sign restriction:

	Unrestricted Model	Restricted Model	Improvement
R^2_{OOS}	-0.01363	-0.0003737	+0.01326
CW Statistic	0.486	2.324	+1.838

Table 6: Impact of Campbell-Thompson Economic Restrictions

After imposing the sign restriction, both performance metrics show dramatic improvement. The out-of-sample R^2 improves from -1.363% to 0.037%. While still slightly negative, it is effectively zero, indicating comparable performance to the historical mean.

The Clark-West statistic increases from 0.486 to 2.324, well above the 5% critical value of 1.645. This allows us to reject the null hypothesis and say that the restricted model provides statistically significant predictive improvement over the random walk.

These results demonstrate that Campbell-Thompson restrictions successfully align statistical estimation with economic theory while enhancing out-of-sample forecast performance.

2.6 Bootstrap Algorithm

While the Clark-West statistic asymptotically follows a standard normal distribution under the null hypothesis in cases with large sample size, finite-sample properties in financial time series can deviate from this theoretical distribution. Bootstrap methods provide more accurate inference by empirically approximating the sampling distribution of the test statistic.

The algorithm proceeds as follows:

1. Start by estimating the model under the null of no predictability:

$$y_t = \alpha + u_{1,t}$$

$$x_t = c + \rho_1 x_{t-1} + \dots + \rho_p x_{t-p} + u_{2,t}$$

with parameters based on the full sample and p selected using the BIC function in R.

2. For each bootstrap replication ($B = 1000$), draw blocks of residuals $(u_t^* = (u_{1,t}^*, u_{2,t}^*)')$ with replacement and generate bootstrap samples:

$$y_t^* = \hat{\alpha} + u_{1,t}^*$$

$$x_t^* = \hat{c} + \hat{\rho}_1 x_{t-1}^* + \dots + \hat{\rho}_p x_{t-p}^* + u_{2,t}^*$$

where $(x_{t-1}^*, \dots, x_{t-p}^*)'$ is a block of consecutive data randomly drawn from the actual data.

3. For each bootstrap sample, estimate both unrestricted and restricted forecasting models and compute the Clark-West statistic.
4. Calculate the Bootstrap p-values as:

$$\text{p-value} = \frac{1}{B} \sum_{j=1}^B I(C\hat{W}_j^* > CW_{obs})$$

for both restricted and unrestricted models, where $I(\cdot)$ is the indicator function.

Table 7 presents the bootstrap p-values for both models, with the corresponding distributions shown in Figure 5:

Model	CW Statistic	Bootstrap p-value	95% Critical Value	Significant at 5%?
Unrestricted	0.486	0.213	1.416	No
Restricted	2.324	0.009	1.373	Yes

Table 7: Bootstrap Analysis Results

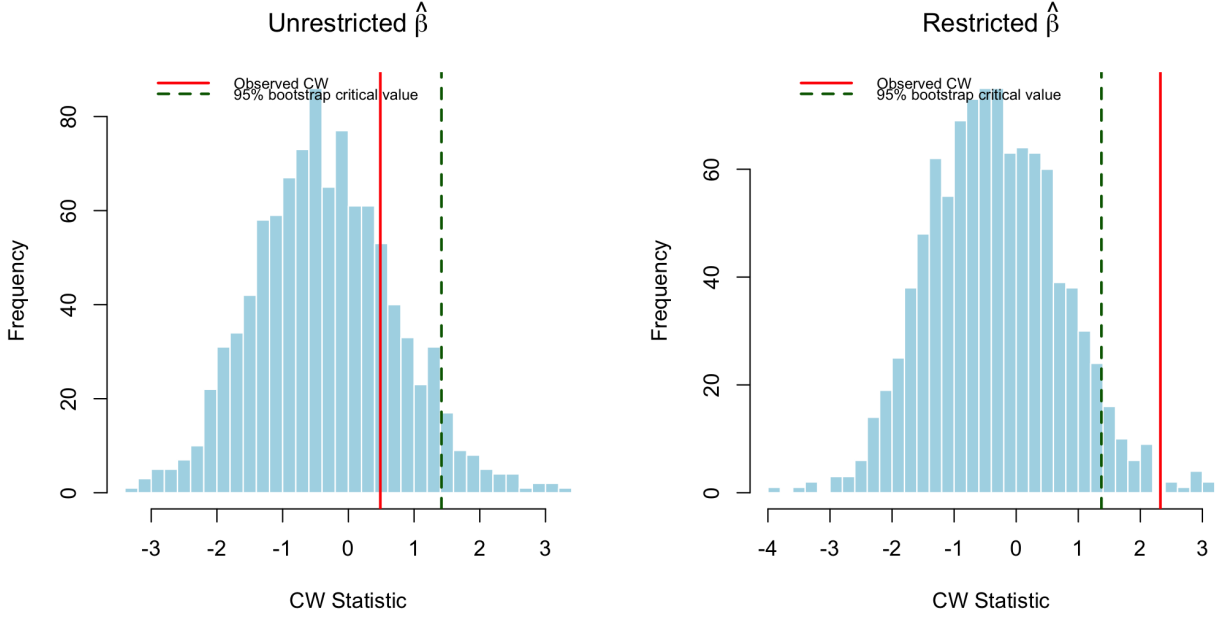


Figure 5: Bootstrap Distributions of Clark-West Statistic

At 5% significance level, the bootstrap p-value for the unrestricted model is 0.213, exceeding the 5% significance threshold, and the observed CW statistic is falls below the critical value. This provides insufficient evidence to reject the null hypothesis of no predictability.

The bootstrap p-value of the restricted model is 0.009, well below the 5% threshold, and the observed CW statistic exceeds the critical value. We therefore reject the null hypothesis with high confidence.

The bootstrap analysis robustly confirms our earlier findings, providing strong evidence that economic sign restrictions are not merely theoretically appealing but statistically essential for extracting reliable predictive signals from the term spread.

2.7 Conclusion

Our analysis shows that the term spread provides statistically significant predicting power for the U.S. stock returns, but only when the Campbell and Thompson sign restriction is imposed. The unrestricted model failed to beat the random walk benchmark ($CW = 0.486$, $p = 0.213$), while the $\hat{\beta} \geq 0$ restriction successfully filtered out noises from structural breaks, particularly during crises like the Great Depression and yielded strong statistical significance ($CW = 2.324$, $p = 0.009$) and substantially improved out-of-sample performance.

While our restricted term spread model achieved statistical significance, its predictive power remains modest $R^2 \approx 0$. Hence, future work should explore other predictors such as the dividend-price ratio, inflation rate, unemployment rate etc. for potential stronger forecasts.

Exercise 3

3.1 Investment Environment and Assumptions

Exercise 2 suggests that the term spread offers limited statistical out-of-sample (OOS) forecasting power for equity returns. From an investment perspective, this is not the end of the story: a predictor can fail to improve mean-squared forecast accuracy and still be valuable if it helps allocate risk across assets when expected premia and risks vary over time. In this section we therefore evaluate the term spread as an input to a disciplined allocation rule, i.e., from the standpoint of a portfolio manager rather than a pure forecaster.

We take the perspective of a US-based investor who rebalances monthly across three investable assets: (i) the US risk-free asset, (ii) the US equity market, and (iii) the UK equity market. UK equity returns are expressed in USD so that currency risk is embedded in the foreign equity leg. Portfolio decisions are formed in real time using a 10-year rolling window: weights at month t use only information available up to t .

To make strategies comparable, portfolio risk is normalised through a constant annual volatility target of $\sigma^* = 10\%$. This volatility-targeting design isolates the economic contribution of the predictive signal from mechanical differences in unconditional risk-taking.

Unless otherwise stated, weights are unconstrained: the strategy may take short positions in risky assets and may borrow/lend at the US short rate (the risk-free weight can therefore be negative). This frictionless benchmark is used to measure the information content of the signal before considering implementation constraints.

Finally, we evaluate realised simple returns for portfolio accounting and compounding, whereas Exercise 2 uses log-returns for regression convenience. When the forecasting step delivers a predicted log return $\ell_{t+1} = \log(1 + r_{t+1})$, we convert it to a simple return via

$$\hat{r}_{t+1} = \exp(\ell_{t+1}) - 1$$

prior to portfolio aggregation. For the purpose of mean–variance allocation, we work with forecasts of excess equity returns, obtained as $\hat{r}_{t+1}^e = \hat{r}_{t+1} - r_t^f$, where r_t^f is the contemporaneous US risk-free rate.

Transaction costs are not imposed in the baseline backtest; instead, we summarise implementability through the break-even cost analysis reported in Section 3.3.

3.2 Portfolio Construction and Allocation Mechanism

Each month we translate the OOS forecasts from Exercise 2 into portfolio positions using a volatility-targeted allocation rule. The practical goal is simple: when the signal implies a stronger expected compensation for bearing equity risk, the portfolio takes more risk exposure; when the signal implies weaker compensation, the portfolio scales risk down and parks more wealth in the risk-free asset. This mapping is what turns a weak forecasting signal into a potentially useful *investment* signal.

Step 1: forecasting inputs. At time t , we apply the predictive-regression forecasting framework of Exercise 2 (10-year rolling window) to construct one-step-ahead forecasts for both US and UK equity returns (UK measured in USD). For the UK leg, we estimate the same regression specification using the corresponding return series and predictor, maintaining full methodological consistency across assets within the allocation problem. UK equity returns are measured in USD, so the forecast for the UK leg reflects both local equity performance and currency movements. We collect the two equity forecasts in a vector $\hat{\mu}_t$ and estimate the conditional covariance matrix of equity returns, $\hat{\Sigma}_t$, from the same rolling window.

Step 2: risk targeting and portfolio weights. We form risky-asset weights so that the portfolio targets a constant annual volatility $\sigma^* = 10\%$. Operationally, this means the total risk taken through the two equity legs is scaled up or down over time, while the relative split between US and UK equities is driven by the forecasted risk-adjusted attractiveness (expected return per unit of risk and their correlation). The remaining weight is allocated to the US risk-free asset, allowing borrowing/lending in the baseline specification.

Economic interpretation of the allocation. Figure 6 reports the realised portfolio weights. Three features are worth highlighting.

First, the strategy is state dependent: equity exposure rises in periods when the signal and the recent covariance estimate jointly imply a higher expected reward-to-risk trade-off, and it contracts when the risk premium signal deteriorates. In other words, the term-spread information is used to *time* overall risk-taking rather than to forecast returns precisely.

Second, the US and UK weights often move in opposite directions. This is consistent with the covariance channel: when the two equity markets offer diversification benefits, the optimiser distributes risk across both; when correlation rises or one market becomes riskier, the strategy concentrates risk in the market with the better forecasted risk-adjusted profile. Because the UK leg is in USD terms, shifts in currency risk can also tilt the UK weight materially.

Third, there are episodes with negative weights (short positions) or a risk-free weight above one

(leveraging the risky legs while borrowing at the short rate). These are not modelling artefacts but a direct implication of running a frictionless mean–variance / risk-targeting benchmark. In practice, such exposures would be moderated by leverage and short-sale constraints; we return to implementability through turnover and break-even transaction costs in Section 3.3.

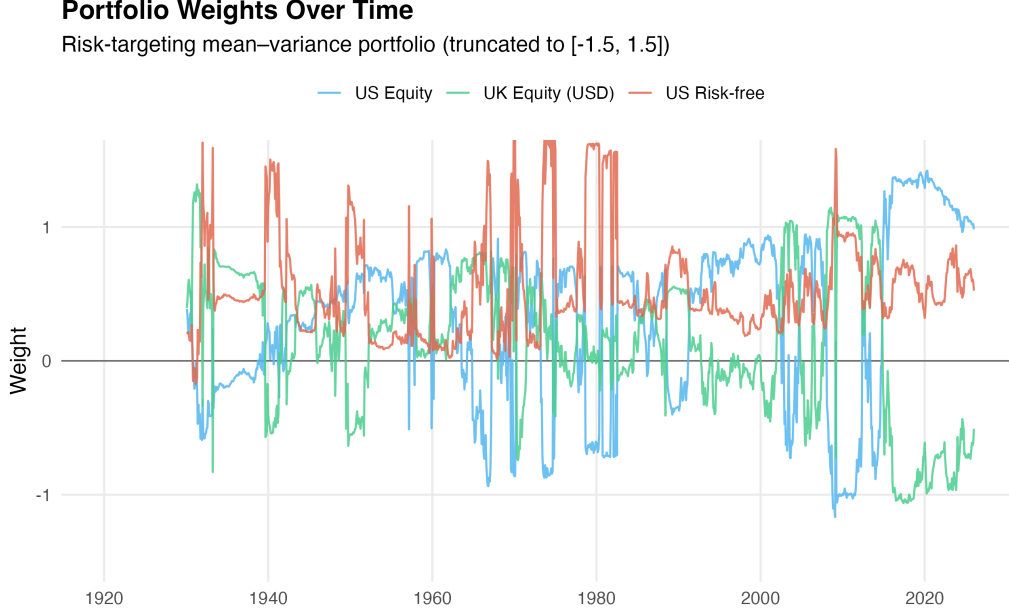


Figure 6: Portfolio weights over time (US risk-free, US equity, UK equity in USD).

3.3 Economic Evaluation and Portfolio Performance

From an investor’s perspective, the question is not whether the term spread produces statistically significant one-step-ahead forecasts (Exercise 2), but whether it can be monetised through a disciplined allocation rule under a fixed risk budget. We therefore evaluate the realised performance of the volatility-targeted strategy and report the required economic metrics: annualised mean and volatility, Sharpe and Sortino ratios, the implied performance fee $\mathcal{P}$ (bps p.a.), and the break-even transaction cost τ (bps/month).

Economic value measures: performance fee and break-even costs. Following Goetzmann, Ingersoll, Spiegel and Welch (2007), we compute the certainty equivalent return (CER) of a strategy as

$$CER = \frac{1}{1 - \delta} \ln \left(E \left[\left(\frac{1 + r_{p,t+1}}{1 + r_{f,t}} \right)^{1-\delta} \right] \right), \quad (19)$$

where δ denotes the coefficient of relative risk aversion. The implied performance fee is then defined as the CER gain of the predictive strategy relative to the benchmark,

$$\mathcal{P} = CER_{MOD} - CER_{BEN}, \quad (20)$$

reported in basis points per annum.

To assess implementability, we follow the lecture break-even transaction cost approach. Let $w_{i,t}$ denote the post-rebalancing portfolio weight on risky asset i at time t , and let $w_{i,t} \left(\frac{1+r_{i,t+1}}{1+r_{p,t+1}} \right)$ be the corresponding pre-rebalancing weight at $t+1$. The period- $t+1$ turnover measure is

$$\tau_{t+1} = \frac{1}{N} \sum_{i=1}^N \left| w_{i,t+1} - w_{i,t} \left(\frac{1+r_{i,t+1}}{1+r_{p,t+1}} \right) \right|, \quad (21)$$

and the break-even proportional trading cost is

$$\tau_{be} = \frac{E \left(r_{p,t+1}^{\text{MOD}} - r_{p,t+1}^{\text{BEN}} \right)}{E \left(\tau_{t+1}^{\text{MOD}} - \tau_{t+1}^{\text{BEN}} \right)}. \quad (22)$$

An investor prefers the predictive strategy (MOD) to the benchmark (BEN) if the per-period proportional trading cost is below τ_{be} .

Summary performance. Table 8 summarises the annualised outcomes. The strategy delivers a mean return of 7.70% p.a. with realised volatility of 11.78% p.a., implying a Sharpe ratio of 0.38 and a Sortino ratio of 0.53 per annum. The implied performance fee is $\mathcal{P} = 59.26$ bps p.a., defined as the CER gain in (20). In addition, the break-even transaction cost is $\tau_{be} = 125.66$ bps/month, defined in (22). In other words, the economic edge depends on how aggressively the allocation trades through time, which we connect to the weights and turnover diagnostics in the surrounding sections.

Table 8: Predictive Strategy: Performance Summary (Annualised)

Metric	Value
Mean (% p.a.)	7.70
Volatility (% p.a.)	11.78
Sharpe Ratio (p.a.)	0.38
Sortino Ratio (p.a.)	0.53
Performance Fee $\mathcal{P}$ (bps p.a.)	59.26
Break-even Cost τ_{be} (bps/month)	125.66

The realised volatility slightly exceeds the 10% target because volatility scaling is based on rolling ex-ante covariance estimates, which are subject to estimation error and discrete monthly rebalancing.

Cumulative wealth and compounding. Figure 7 plots cumulative portfolio returns for the predictive strategy and the historical-mean benchmark on a logarithmic scale. The predictive strategy achieves a persistently higher growth path over the sample, leading to a substantially larger terminal wealth. Importantly, the divergence between the two strategies emerges gradually rather than being driven

by a single extreme episode. The near-parallel yet consistently higher trajectory of the predictive strategy indicates a modest but sustained improvement in average performance, which compounds into economically meaningful differences in long-run wealth.

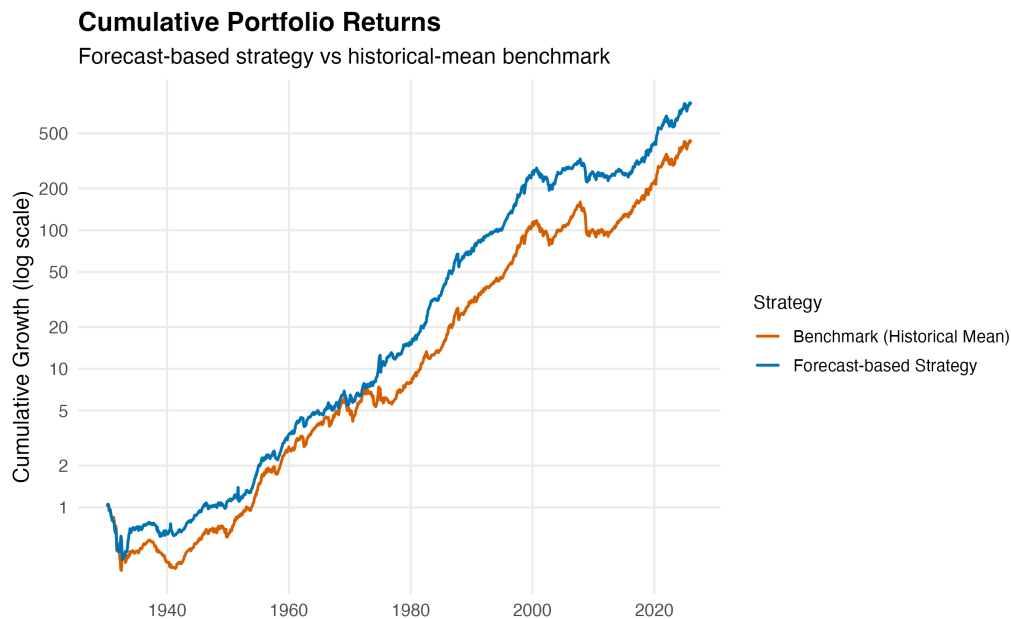


Figure 7: Cumulative Returns: Predictive Strategy vs Benchmark

Downside risk and investor experience. Cumulative performance is only one side of the investment experience. A strategy that outperforms by taking larger crashes is typically unattractive to real investors. Figure 8 therefore compares drawdowns. The predictive strategy does not avoid major market stress episodes, but it generally does not exhibit structurally deeper drawdowns than the benchmark. In several stressed intervals, drawdowns appear modestly contained, which is consistent with the mechanism of a term-spread signal: when the yield curve flattens/inverts, expected compensation for equity risk tends to deteriorate and a volatility-targeted allocator can endogenously scale back risk.

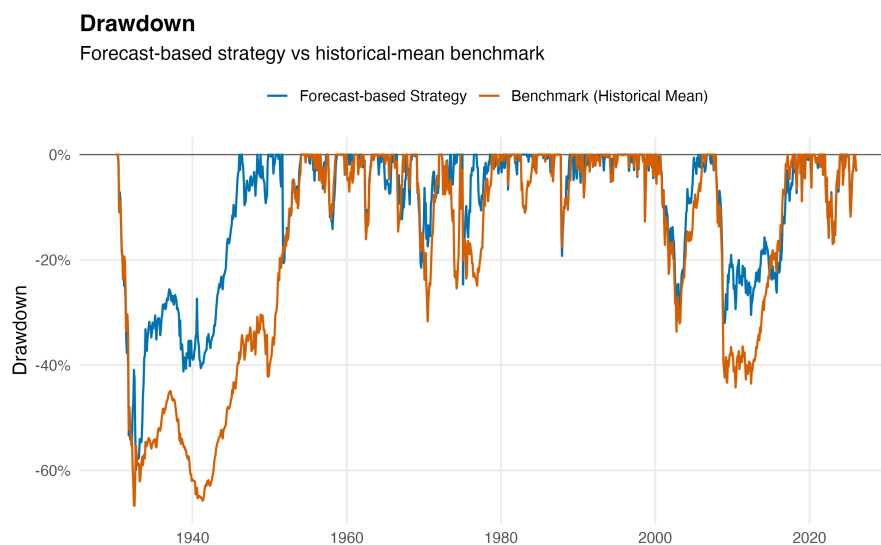


Figure 8: Drawdown: Predictive Strategy vs Benchmark

Time variation in risk-adjusted efficiency (1-year rolling Sharpe). While the full-sample Sharpe ratio is positive, investors care about stability. Figure 9 plots the one-year rolling Sharpe ratio. Two features stand out. First, short-horizon Sharpe ratios are highly time-varying and frequently cross zero for both the benchmark and the predictive strategy. This highlights that, at the monthly horizon, the realised payoff to timing signals is noisy even when the long-run edge is positive. Second, the predictive strategy tends to improve relative efficiency during some macro regimes, while underperforming in others—consistent with the view that the term spread acts as a state variable whose usefulness depends on the business-cycle and risk-premium environment.

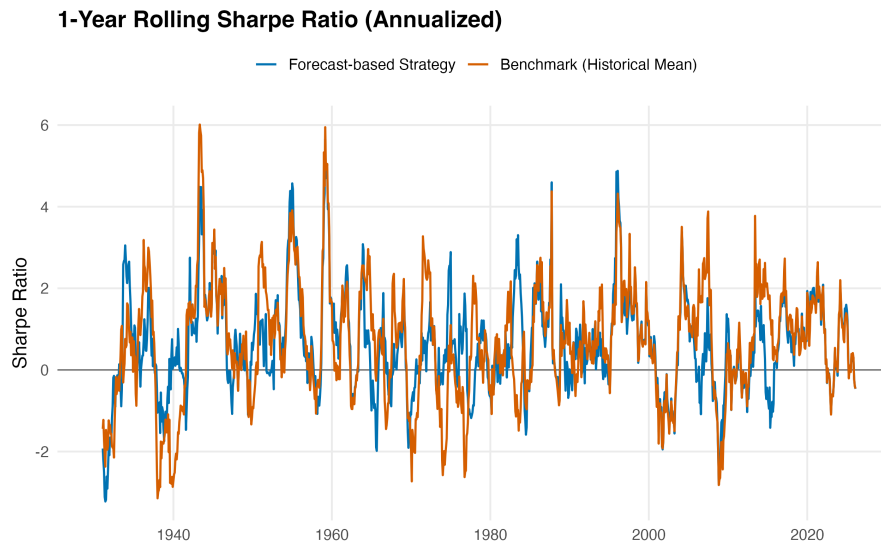


Figure 9: 1-Year Rolling Sharpe Ratio (Annualised)

Structural performance over longer horizons (5-year rolling Sharpe). Short-window Sharpe ratios can be dominated by transitory shocks. To assess whether the signal adds value in a more persistent sense, Figure 10 reports five-year rolling Sharpe ratios. Relative to the one-year metric, the five-year Sharpe is smoother and more interpretable as a measure of medium-term efficiency. The predictive strategy shows periods of sustained improvement in risk-adjusted returns, but not uniformly. This pattern aligns with the broader message of Exercise 2: the term spread is not a reliable point forecaster each month, yet it can still help allocate risk more effectively over longer horizons when risk premia are meaningfully time-varying.

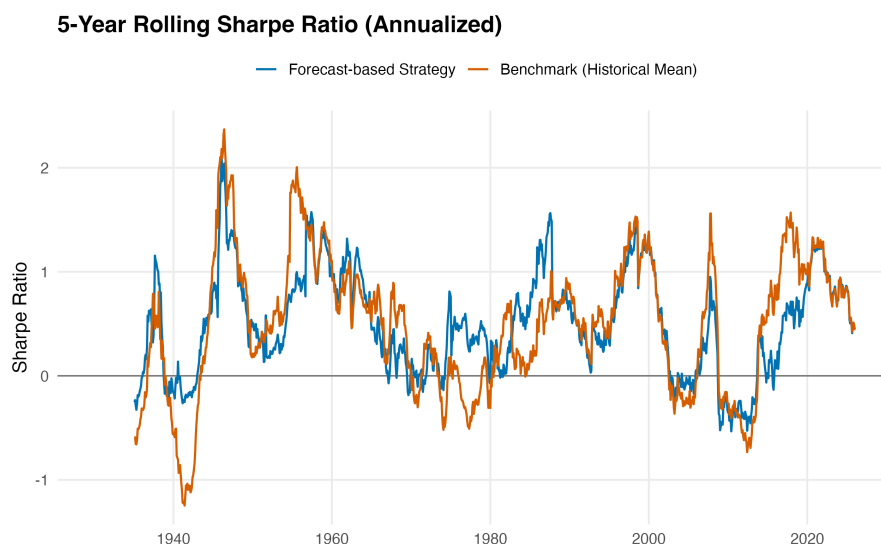


Figure 10: 5-Year Rolling Sharpe Ratio (Annualised)

Interpretation and implementability. Taken together, the evidence suggests that economic value can arise even when statistical OOS tests are weak: what matters is how forecast information is mapped into portfolio risk. That said, the magnitude of τ should be interpreted alongside trading intensity. If the strategy requires frequent and large reallocations (as suggested by the weight dynamics in the portfolio-weight plot), realised net performance could be sensitive to implementation frictions. In the next section, we therefore link performance to the dynamics of portfolio weights and discuss practical constraints such as leverage and trading costs.

3.4 Implementation, Risk Controls and Economic Interpretation

The previous section shows that the predictive allocation generates economically meaningful improvements in long-run wealth and medium-horizon risk-adjusted performance. The remaining question is whether such gains are realistically attainable once implementation frictions and institutional constraints are taken into account.

Leverage and borrowing assumptions. The volatility-targeting rule implies that portfolio weights can occasionally exceed unity in absolute value, especially when forecasted excess returns are high relative to estimated variance. This effectively introduces leverage. In our baseline implementation, we allow borrowing at the US risk-free rate and short positions in risky assets. This assumption is standard in mean–variance allocation but should be interpreted carefully.

From an institutional perspective, leverage amplifies both gains and losses. The fact that the realised volatility remains close to the 10% target indicates that leverage is used primarily as a scaling device rather than as an uncontrolled risk amplifier. However, if leverage were capped (e.g., no borrowing or

no short-selling), the strategy's return differential would likely be attenuated. In that sense, part of the economic value derives from dynamic risk scaling rather than pure directional timing.

Trading intensity and cost sensitivity. Portfolio weights fluctuate materially over time (as seen in the weight dynamics of Section 3.2), implying non-trivial turnover. The reported break-even transaction cost of $\tau = 125.66$ bps per month provides a benchmark for implementability.

For highly liquid instruments such as index futures or ETFs tracking US and UK equity markets, actual transaction costs are typically well below this threshold, particularly for institutional investors. This suggests that, in principle, the strategy remains viable after reasonable trading costs. Nevertheless, in periods where the signal reverses sharply, turnover spikes could temporarily erode performance.

Economic mechanism. The term spread can be interpreted as a proxy for the state of the business cycle and time-varying risk premia. A steep yield curve is often associated with stronger expected growth and higher equity compensation, while a flat or inverted curve signals tightening financial conditions and lower future excess returns.

The strategy does not forecast crises perfectly, nor does it avoid all drawdowns. Instead, it adjusts the scale of equity exposure as the macroeconomic state evolves. The performance improvements therefore arise from better risk allocation across regimes rather than precise return timing. This distinction reconciles the weak statistical evidence in Exercise 2 with the positive economic value observed in portfolio outcomes.

Limitations and robustness. Several caveats are worth noting. First, parameter estimation relies on rolling-window OLS, which may be sensitive to structural breaks. Second, the signal's effectiveness varies across subperiods, as evidenced by the rolling Sharpe ratios. Third, leverage assumptions and rebalancing frequency materially affect realised performance.

Future extensions could incorporate: (i) explicit leverage constraints, (ii) transaction-cost penalties in the allocation rule, (iii) multi-predictor combinations rather than a single term-spread signal.

Overall assessment. In summary, while the term spread does not deliver strong out-of-sample forecast accuracy in a purely statistical sense, translating the signal into a disciplined volatility-managed allocation produces economically meaningful improvements in long-run wealth and medium-horizon efficiency.

For an institutional investor with access to liquid index instruments and moderate leverage capacity, the strategy appears implementable and economically defensible. Its value lies not in precise market timing but in dynamic risk management across macroeconomic regimes.

Appendix (references)

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